USHER: User-Specific Hallucination Estimation from Reactions in LLM Agents

Anonymous submission

Abstract

An agent serving a user must act on facts about them it was never told, such as whether a scan is safe today, and nothing external can check its confident guesses. We show that sampling-consistency detectors cannot tell users apart within a conversation, and bound how far they rise above chance by how often resamples disagree with wrong and with right answers. We propose USHER (User-Specific Hallucination Estimation from Reactions): the agent states each assumption as a commitment with a confidence, the user's implicit reactions are recorded, and a posterior turns both into per-commitment and per-session risk. In a pre-registered evaluation with five LLM agents, SelfCheck-GPT and semantic entropy score 0.551 and 0.495 AUROC. An intake-aware posterior, registered after the first results and run once, reaches 0.825, beats a supervised detector on the same inputs by 0.089 (one given its structure matches it), and identifies which user-specific commitment is wrong at 0.797 AUROC, where resample disagreement reaches 0.541. Exact Bayes predictors show the reactions carry information beyond population risk that resampling ignores. One question about the posterior's riskiest commitment cuts user-specific errors by 43%, a random one by 9%. We report every registered outcome, including failures.

Keywords: LLM agents; hallucination detection; uncertainty; clarification; implicit feedback.

1 Introduction

An agent that schedules imaging appointments must decide whether a patient may have a contrast scan today. The intake form says the patient had a recent treatment, but not when. A careful agent would say it does not know; many instead answer "safe, $p = 0.85$" and proceed. We call this a *user-specific hallucination*: a confident assertion about the current user whose truth depends on something the user never disclosed. It differs from the factual hallucinations that dominate the literature (Ji et al., 2023): there is no document or search engine to check it against.

The leading black-box detectors resample the model and flag answers it does not reproduce (Manakul et al., 2023; Kuhn et al., 2023; Farquhar et al., 2024). That presumes hallucinations are stochastic. Fabricated user facts need not be: an underdetermined question maps the same context to the same guess every time. More fundamentally, two users who share a conversation (the same intake answers and, where the agent is shown them, the same reactions) are indistinguishable to the model, so nothing computed from the model and its context can tell which of them it is wrong about (Theorem 1). Context can still reveal *population* risk, how often a commitment is wrong among users with the same intake, and we show that a well-calibrated context-only predictor can exploit it. Sampling-consistency scores of a self-consistent agent do not even do that (Corollary 1).

The way out is to look at the user rather than the model. In USHER the agent must externalize each assumption as a *commitment* $(\text{attribute}, \text{value}, p)$, and the user's reaction (accepting it, contradicting it, or ignoring it) is recorded. Reactions are noisy and partial, but they depend on the truth for *this* user, which no resample does: under a simple model the posterior of the hallucination event uses them exactly (Proposition 1), and, with the agent's trajectory held fixed, less noisy reactions cannot hurt (Theorem 2). Our contributions:

1. **Why resampling fails, and by how much.** No detector computed from the model and its context (resampling, self-verification, hidden-state probes) can tell which of two users sharing a conversation it is wrong about (Theorem 1), and a dispersion score's AUROC lies within $\frac{1}{2}e_0e_1$ of $\frac{1}{2} + \frac{1}{2}(e_1 - e_0)$, where e_1 and e_0 are how often resamples disagree with wrong and with right commitments (Corollary 1). On fresh sessions from five agents, all three resamples agree on 50–91% of the commitments whose truth depends on undisclosed facts, resample disagreement predicts their errors at 0.501–0.555 AUROC, and SelfCheck-GPT and semantic entropy score 0.551 and 0.495 on user-specific hallucination.

2. **A protocol that makes these errors detectable.** In USHER the agent states each assumption as a commitment with a confidence and the user's implicit reaction is recorded. Detectors on these observables beat both sampling detectors by +.112 to +.219 AUROC pooled, and the user's reactions add detectable signal beyond population base rates (+.070; significant for 5 of 5 agents).

3. **A posterior that says which claim is wrong, and what to ask.** Our label-event posterior is exact under a conditional-independence model, and with a symmetric channel its priors are identified from reactions

alone up to one noise parameter (Proposition 2). Its intake-aware form, V7, learns which commitments the intake leaves to the user. Designed on development data after the registered results and registered before its first run, it reaches 0.825 AUROC and beats the supervised detector on the same raw inputs (registered H8: +.089); a classifier handed its structure matches it, and its gain is smallest on Mistral-24B. The user's reactions add +.083 to it. Per commitment it identifies the wrong user-specific ones at 0.797 AUROC with calibration error 0.081, against 0.752 for the commitment's own reaction and 0.541 for resample disagreement. Asked about its riskiest commitment, the agent leaves 43% fewer wrong confident user-specific commitments, against 9% for a random question (registered H9).

4. **Where the information is.** Exact Bayes predictors in a simulator with known truth show that sampling fails not for lack of information: the population risk of the commitments alone reaches 0.781–0.899 AUROC, the reactions add +.083 beyond it, and nearly all of that reaches the agent's transcript.

5. **A pre-registered evaluation, fully reported.** After a blind internal audit of our development results we fixed a protocol bug, registered every comparison and its wording, and generated fresh sessions with five agents from four families and two domains. We report every registered outcome, including those against our first posterior, a simulated patient that failed its quality gate, and a second one whose weaker reactions erase V7's advantage.

2 Related Work

Hallucination detection. Sampling-based detectors measure agreement across resamples (Manakul et al., 2023; Kuhn et al., 2023; Farquhar et al., 2024); semantic entropy is explicitly scoped to stochastic confabulations, not consistent errors (Farquhar et al., 2024). Learned confidence models from a model's generations (Ulmer et al., 2024), white-box probes (Azaria and Mitchell, 2023; Chen et al., 2024; Kossen et al., 2024), and verification by the model's own questions (Dhuliawala et al., 2024) or by search (Wei et al., 2024) use nothing about the user beyond the context, so Theorem 1 applies to them. Consistent errors are documented in snowballing (Zhang et al., 2024) and self-contradiction studies (Mündler et al., 2024), and calibrated models must hallucinate on facts absent from training (Kalai and Vempala, 2024); our setting is the per-user analogue. Verbalized confidence (Kadavath et al., 2022; Tian et al., 2023; Xiong et al., 2024) is a commitment-level signal we include as a baseline, and iterative prompting can separate epistemic from aleatoric uncertainty (Abbasi Yadkori et al., 2024), still from the model given its context.

Hallucination in agents. MIRAGE-Bench (Zhang et al., 2025) taxonomizes agent actions unfaithful to instructions, history or observations; CAR-bench (Kirmayr et al., 2026) tests whether agents admit missing information with a simulated user; AgentHallu (Liu et al., 2026) localizes the step that introduced an error. In multiagent systems a commitment is a public, checkable obligation of one agent to another (Singh, 1999), and a probabilistic commitment binds an agent to achieve an outcome with a stated probability (Zhang et al., 2016); ours are probabilistic assertions about the counterpart's private state, checked against its reactions. Dialogue models hallucinate about their partner as well as about the world (Dziri et al., 2022). We study detection when the missing information is the user's own.

Clarification. When to ask has been framed through expected value of perfect information (Rao and Daumé III, 2018; Suri et al., 2025), ambiguity detection (Zhang and Choi, 2025), conformal prediction (Ren et al., 2023), and information-gain rewards on τ-bench (Deng et al., 2026; Yao et al., 2024). In clinical dialogue, MediQ (Li et al., 2024) and Uncertainty of Thoughts (Hu et al., 2024) ask a simulated patient questions to reduce the doctor model's uncertainty. These reason about the agent's uncertainty before it acts; we use post-hoc commitment risk to choose which assertions to confirm, and measure the effect with the agent re-reasoning.

Implicit and noisy feedback. Implicit signals such as clicks are informative but biased (Joachims et al., 2005). Conversation partners ground what they assert through the other's acceptance (Clark and Brennan, 1991); spoken dialogue systems confirm hypotheses implicitly and update beliefs from the user's reaction (Bohus and Rudnicky, 2005; Williams and Young, 2007), and chatbots can learn from their partner's satisfaction signals (Hancock et al., 2019); our reactions play the same role for asserted user facts. The general channel of our posterior, with error-dependent contradiction and confirmation rates, has the form of the guess and slip parameters of knowledge tracing (Corbett and Anderson, 1994). Our estimation step inverts a noise channel, as in learning from noisy labels (Angluin and Laird, 1988; Natarajan et al., 2013; Scott et al., 2013) and annotator models (Dawid and Skene, 1979). Agents that revise under user pushback raise sycophancy concerns (Sharma et al., 2024); our simulated user follows a long line of user simulation for dialogue (Schatzmann et al., 2006).

3 Problem, Protocol, and Labels

Definition 1 (User-specific hallucination)**.** *A* commitment *asserts a value v for an attribute a of the current user with confidence $p \in [0,1]$. It is* user-dependent *if its true value is not determined by what the user has disclosed, and it is a user-specific hallucination if it is user-dependent, confident ($p \geq \tau$), and wrong.*

Protocol. An USHER session has four stages. (i) *Intake*: the user answers a few coarse questions; the agent sees only these answers. (ii) *Commitment*: at each turn the agent emits one commitment in a fixed format, `[attribute]: value, p=confidence`. (iii) *Reaction*: the user's implicit reaction $F_t \in \{-1, 0, +1\}$ (contradicts, no reaction, accepts) is recorded and shown to the agent for its next six turns. (iv) *Detection and intervention*: a detector scores the session and its commitments; high-risk commitments trigger a confirmation question.

Labels. With confident set $H = \{t : p_t \geq \tau\}$, $\tau = 0.6$, the *composite* label L (Eq. (1)) marks a session in which at least half of the confident commitments are wrong ($\rho = \frac{1}{2}$), and is 0 if there are none. It mixes two kinds of error. Some attributes are determined by the intake through rules the agent was never shown (e.g. contrast is needed iff urgency is medium or high); errors there are guesses about the clinic, not about the user. The *user-specific* label L_U applies the same rule to the confident commitments that are user-dependent in that session, and is 0 when there are none. Whether a commitment is user-dependent is decided per session by enumerating the hidden profile's support: `scan_safe_now`, for example, depends on days since treatment only when a treatment was recent; `needs_contrast` is fixed by urgency, so a wrong "no" counts in L but never in L_U. L_U is our primary label; L is reported as secondary.

4 What Resampling and Feedback Can Reveal

Setup. A user u is drawn from a population. At turn t the agent conditions on a context X_t: the coarse intake $\phi(u)$, the question, and the interaction history, which in USHER displays earlier reactions; X denotes $(X_1, \ldots, X_T)$. It emits a commitment $c_t = (a_t, v_t, p_t)$ (attribute, value, reported confidence) from its policy $\pi(\cdot \mid X_t)$. Apart from the truth and the reactions, nothing depends on u except through X. The truth $y_a(u)$ is private to the user; the commitment is wrong if $w_t = \mathbf{1}[v_t \neq y_{a_t}(u)]$. With confident set $H = \{t : p_t \geq \tau\}$ and error fraction ρ, a session hallucinates iff

$$L = \mathbf{1}\Big[\,|H| \geq 1 \text{ and } \textstyle\sum_{t \in H} w_t \;\geq\; \lceil \rho\,|H| \rceil\Big]. \qquad (1)$$

The results below hold for any binary label determined by (C, u): a single w_t, or L_A, Eq. (1) with H replaced by $H_A = \{t \in H : a_t \in A\}$ for an attribute set A determined by X (for the user-specific label L_U, the attributes the intake does not determine). We write L.

A *context-and-sample detector* is any $S = h(X, C, Z, \xi)$ with ξ independent randomness and Z drawn, given (u, X, C), from a kernel $K(\cdot \mid X, C)$ that does not involve u: resamples at any temperature, token log-probabilities, hidden states. A *feedback detector* also sees reactions $F_t \in \{-1, 0, +1\}$ (contradicts, none, accepts). By the Neyman–Pearson (NP) lemma (Neyman and Pearson, 1933), among all, possibly randomized, detectors that are functions of an observation O, thresholding $P(L = 1 \mid O)$ (randomized at ties) has the highest true-positive rate at every false-positive rate, hence the highest AUROC; ties count $\frac{1}{2}$. Theorem 1 is the standard data-processing consequence of NP (Neyman and Pearson, 1933; Blackwell, 1953), stated for this setting.

Theorem 1 (Self-consistency bound)**.** *Let $\eta_X = P(L = 1 \mid X, C)$ and S be a context-and-sample detector. Given (X, C), S is independent of u and L, so within a context it ranks hallucinating above clean sessions with probability exactly $\frac{1}{2}$. Across contexts, its ROC curve lies pointwise below that of the Bayes predictor η_X.*

Proof. Given (u, X, C), S depends only on (Z, ξ), whose law does not involve u, and L is a function of (C, u). So the law of S given (X, C) does not depend on L, $P(L = 1 \mid X, C, S) = \eta_X$, and NP applies to $O = (X, C, S)$. □

Since X contains the reactions the agent was shown, η_X is not feedback-free, and detectors built from (X, C), such as context-only classifiers, can reflect them and exploit base rates across contexts. Resampling cannot add to (X, C) or separate users who share a context; a feedback detector reads the reactions directly, including the last. For dispersion scores, more is true:

Corollary 1 (Dispersion under systematic fabrication)**.** *Let S be a score of the commitments and their k resamples (agreement rate, semantic entropy, SelfCheck-style inconsistency) that takes a fixed value $s^\star$ whenever every resample equals its commitment, and $e_\ell = P(S \neq s^\star \mid L = \ell)$. Then*

$$\big|\mathrm{AUROC}(S) - \tfrac{1}{2}\big| \;\leq\; \tfrac{1}{2}\big(1 - (1 - e_0)(1 - e_1)\big) \;\leq\; \tfrac{1}{2}(e_0 + e_1).$$

If $s^\star$ is the least suspicious value of S, then $\big|\mathrm{AUROC}(S) - \frac{1}{2} - \frac{1}{2}(e_1 - e_0)\big| \leq \frac{1}{2} e_0 e_1$. If every turn's resampling distribution is a point mass at the committed value, $\mathrm{AUROC}(S) = \frac{1}{2}$ however often the commitments are wrong.

Proof. For independent positive and negative draws, both scores equal $s^\star$ with probability $(1 - e_0)(1 - e_1)$; such pairs are ties and contribute exactly $\frac{1}{2}$. All other pairs contribute a value in $[0, 1]$. If $s^\star$ is least suspicious, a pair where only the positive (negative) differs from $s^\star$ contributes 1 (0); only pairs where both differ, with probability $e_0 e_1$, are free. □

Table 2 checks the corollary on every agent.

Theorem 2 is Blackwell's garbling theorem (Blackwell, 1953) for a symmetric channel. Let F^ε be feedback generated for the detector: given (u, X, C), which F_t^ε are 0 has a law depending only on (X, C), and each other F_t^ε is $1 - 2w_t$ with its sign flipped independently with probability ε. Let $\eta_\varepsilon = P(L = 1 \mid X, C, F^\varepsilon)$ ($\varepsilon = 0$ is perfect feedback).

Theorem 2 (Less noisy feedback cannot hurt)**.** *Fix the joint law of (u, X, C); in particular, it does not depend on*

ε. *For $0 \le \varepsilon \le \varepsilon' \le \frac{1}{2}$ the ROC curve of η_ε lies pointwise above that of $\eta_{\varepsilon'}$, which lies above that of η_X, and $\eta_{1/2} = \eta_X$ almost surely.*

Proof. If $\varepsilon' = \varepsilon$ there is nothing to show. Otherwise $\varepsilon < \frac{1}{2}$, and flipping each nonzero F_t^ε independently with probability $\delta = (\varepsilon' - \varepsilon)/(1-2\varepsilon) \in [0, \frac{1}{2}]$ yields flip rate $\varepsilon + \delta(1-2\varepsilon) = \varepsilon'$, hence the joint law of $(u, X, C, F^{\varepsilon'})$. So every detector using $F^{\varepsilon'}$ is a randomized detector using F^ε, and NP gives the first inequality; ignoring F gives the second. At $\varepsilon = \frac{1}{2}$, F^ε is independent of u given (X, C): whether a reaction occurs depends only on (X, C), and every observed sign is a fair coin. □

The premise is essential: if the agent conditions on the same reactions, the law of (X, C) changes with ε and monotonicity can fail (a two-turn counterexample is in the repository). The theorem covers reactions regenerated for the detector with the agent's trajectories fixed.

The label-event posterior. Let $\kappa_t = (a_t, v_t, b(p_t))$ be the detector-visible *cell*, where $b(\cdot)$ buckets confidence, and $\mathscr{H}_t = (X_t, c_{1:t}, F_{<t}, w_{<t})$ the history up to the t-th commitment with the earlier errors. (A1) Given $\mathscr{H}_t$, $w_t \sim \text{Bernoulli}(\pi_{\kappa_t})$. (A2) Given $(\mathscr{H}_t, w_t)$, $F_t = 0$ with a probability that depends only on a_t, and otherwise $F_t = 1 - 2w_t$ with its sign flipped with probability ε. Up to a factor free of w_t, the likelihood of F_t is $\ell_{w_t}(F_t)$ with $\ell_w(0) = 1$, $\ell_1(-1) = \ell_0(+1) = 1 - \varepsilon$ and $\ell_1(+1) = \ell_0(-1) = \varepsilon$. (A1) excludes error cascades and intake information beyond the cell, but lets the agent choose cells after seeing feedback.

Proposition 1 (Exact posterior)**.** *Under (A1)–(A2), $P(w_{1:T} \mid X, C, F) = \prod_t q_t^{w_t}(1-q_t)^{1-w_t}$ with $q_t = \pi_{\kappa_t}\ell_1(F_t) / \big(\pi_{\kappa_t}\ell_1(F_t) + (1-\pi_{\kappa_t})\ell_0(F_t)\big)$. Hence $\eta_\varepsilon = P(L = 1 \mid X, C, F)$ is 0 if $H = \varnothing$ and otherwise the Poisson-binomial tail $P\big(\sum_{t\in H} B_t \ge \lceil \rho |H| \rceil\big)$, $B_t \sim \text{Bernoulli}(q_t)$ independent, computable in $O(|H|^2)$. It is Bayes-optimal under (A1)–(A2) with the true priors and channel; the implementation plugs in estimates.*

Proof. Write the joint law in the order $c_1, w_1, F_1, c_2, \ldots$. The policy and the questions see u only through X, so $p(X, C, F, w) = A(X, C, F) \prod_t \pi_{\kappa_t}^{w_t}(1-\pi_{\kappa_t})^{1-w_t}\ell_{w_t}(F_t)$, where A collects the intake prior, the questions, the policy and the w-free factors of (A2). Normalizing over $w \in \{0,1\}^T$ gives independent Bernoulli(q_t). H is fixed by C, L depends on w only through $\sum_{t\in H} w_t$, and NP gives optimality. □

The same proof gives the posterior of L_A (the tail over H_A), and holds for any channel $P(F_t \mid \mathscr{H}_t, w_t) = \ell_{w_t}(F_t)$ that may depend on a_t, as in the general-channel detector. Under (A1) this user's reactions enter only through $\ell_1(F_t)/\ell_0(F_t)$; setting these to 1 leaves the population base rates π_κ (the prior-only ablation). Cascades violate (A1); a simple cascade model did not help on development data (repository).

Proposition 2 (Priors are identified up to one scale)**.** *Assume (A2), let $\pi_\kappa = P(w_t = 1 \mid \kappa_t = \kappa)$, and let $\hat{r}_\kappa$ be the fraction of the n_κ reacted-to commitments in cell κ that were contradicted, with independent errors given n_κ. Then $r_\kappa = \varepsilon + (1-2\varepsilon)\pi_\kappa$; for known $\varepsilon < \frac{1}{2}$, $\hat{\pi}_\kappa = (\hat{r}_\kappa - \varepsilon)/(1-2\varepsilon)$ needs no labels and, with probability at least $1-\delta$, $|\hat{\pi}_\kappa - \pi_\kappa| \le (1-2\varepsilon)^{-1}\sqrt{\log(2/\delta)/(2n_\kappa)}$ (Hoeffding, 1963). Cells that never receive reactions are not identified, and if ε is unknown every $\varepsilon' \le \min_\kappa \min(r_\kappa, 1-r_\kappa)$ fits equally well.*

We fix the scale by training-split label likelihood; unidentified cells get the pooled bucket prior, and estimates are clipped to $[0.01, 0.99]$ and shrunk toward coarser cells.

5 Detectors

The protocol yields two observables per turn, a commitment and a reaction. We read them with two kinds of *USHER detectors*: a label-event posterior, which also gives per-commitment risk, and supervised classifiers. The sampling baselines read only the model and its context. Table 1 lists what each one sees.

USHER posterior. The score is a plug-in estimate of the posterior of Proposition 1. Each confident commitment falls in a cell $\kappa = (\text{attribute}, \text{value}, \text{confidence bucket})$ (buckets $[0.6, 0.65)$, $[0.65, 0.75)$, $[0.75, 1]$); cell priors come from training reactions by the method of moments (Proposition 2), shrunk toward attribute- and bucket-level estimates ($k = 5$). The registered primary detector, V6, uses a general channel: contradiction rates $r_w = P(F = -1 \mid w)$ chosen on training labels ($r_1 > r_0 + 0.1$), priors $\pi = (\text{rate} - r_0)/(r_1 - r_0)$, and confirmation rates fit by least squares; V3 is the symmetric special case. For L_U the event is restricted to attributes that depend on a hidden field anywhere in the schema, which the detector can know without the clinic's rules; V6 is therefore the posterior of a proxy event (L_A in Section 4). Whether a commitment is wrong also depends on the intake, which the cell omits, so (A1) fails and the plug-in is misspecified. V6−prior keeps the fitted priors and ignores the test user's reactions.

Intake-aware posterior (V7). V7 repairs both mismatches and nothing else. Its cells add the intake answers each attribute depends on in the schema, shrunk toward V6's cell (errors still share hidden facts across attributes, so (A1) remains an approximation). Its label event is learned: a commitment whose truth the intake fixes is wrong for every user with that intake or for none, a rule error rather than a user-specific one. Starting from the schema-level set, V7 removes an (attribute, intake) cell from the event, or returns it, when that lowers the training log-loss, alternating with the channel fit; for L, whose event is every confident commitment, only the cells change. On development data the intake cells alone add nothing on L_U; the gain there comes from the learned event. V7 was designed after the registered results were known, tuned on the development corpora only (twelve variants, all logged), and frozen and

Table 1: What each detector sees. Int.: intake answers; Com.: commitments and stated confidences; Rea.: reactions; Res.: resamples or verification queries; Sch.: the schema's dependency structure; Rul.: the simulator's rules and hidden-fact prior; Lab.: training labels. Oracle rows are not deployable.

Detector	Int.	Com.	Rea.	Res.	Sch.	Rul.	Lab.
SelfCheckGPT, semantic entropy	·	✓	·	✓	✓	·	·
CoVe	✓	✓	✓	✓	·	·	·
LLM judge	✓	✓	✓	·	·	·	·
Stated confidence	·	✓	·	·	·	·	✓
Contradiction fractions	·	✓	✓	·	·	·	·
ctx-LR / ctx-LR+F	·	✓	+F	·	·	·	✓
Intake-LR+F	✓	✓	✓	·	·	·	✓
Structured LR+F	✓	✓	✓	·	✓	·	✓
V6	·	✓	✓	·	✓	·	✓
V7	✓	✓	✓	·	✓	·	✓
$\bar{\eta}$ (oracle)	✓	✓	·	·	✓	✓	·
η_X, η_F (oracle)	✓	✓	✓	·	✓	✓	·
V6-/LR-session (oracle)	✓	✓	✓	·	✓	✓	✓

registered (Amendment 6) before its first run on confirmatory data, when the scores of its H8 comparator (intake-LR+F, below) on four agents were already known.

Supervised USHER classifiers. *ctx-LR* is an L_2 logistic regression ($C = 0.1$, fixed) on counts of confident commitments per (attribute, value) cell, $|H|$, the number of turns and mean confidence, with no reactions and no samples; *ctx-LR+F* adds counts per (attribute, value, reaction) cell. Neither sees the intake answers; "context-only" in the registered wording of H3 means no reactions and no samples. Both need the structured commitments that USHER elicits. Two feedback statistics (fraction of contradicted confident commitments, fraction of contradicted turns) are the simplest USHER detectors.

Baselines. SelfCheckGPT and semantic entropy on $k = 3$ resamples per turn at temperature 0.9 (exact-match clustering, adequate for categorical answers; for L_U computed on turns whose attribute is in the schema-level set above); Chain-of-Verification with the agent as its own verifier (Dhuliawala et al., 2024); mean stated (verbalized) confidence, sign chosen on the training split. Detectors on a graph of the commitments and their reactions (a logistic regression on 14 graph statistics, a two-layer GCN (Kipf and Welling, 2017), and the best on test of five block-model graphon likelihoods (Lovász, 2012)) score 0.41–0.66 on L_U, below V7 on every agent; we omit them from the tables.

6 Experimental Design

Environment. A simulated imaging clinic. Five coarse intake answers are shown to the agent; four hidden fields (days since treatment, allergy severity, kidney function, anxiety) are not. Each session asks for 6–12 of 12 attributes in dependency order. Six attributes can depend on a hidden field, but only for some intakes (a seventh, `transport_assistance`, depends on one in the schema but is always fixed by the intake). Ground truth is a deterministic function of intake and hidden fields; the hidden fields are independent draws from a fixed prior. The agent is a real LLM; only the patient is simulated. A second domain, a travel-booking assistant, has the same structure with different content: five intake answers, four undisclosed traveller facts (such as mobility and dietary needs), and twelve booking attributes, six of which can depend on them. It was specified and registered before any of its data existed.

Reactions. In the main corpora the reaction to a commitment is its correctness bit flipped with probability $\varepsilon = 0.25$, absent for two attributes the patient never reacts to. In the *LLM-patient* corpus a second model (Llama-3.1-8B) role-plays the patient: it is given its own history but not the clinic's decision, one of three personas (attentive, trusting, anxious), and the agent's commitment as a hedged plain-language sentence, and replies in one or two sentences, which the agent reads verbatim. A labeller model (in the first version, the same 8B model) that sees only the sentence and the reply labels it as agreement, contradiction or neither. An *informed* variant also tells the patient the clinic's decision. A registered gate guards this corpus: a 100-session pilot runs first, and the full 600 sessions run only if contradictions are at least 1.5 times as frequent after wrong commitments as after correct ones.

Development and confirmation. Method development used two development corpora (Llama-3.3-70B, 600 sessions; Llama-3.1-8B, 200). A blind internal audit of these results found that the agent had been shown a corrupted feedback display, that resamples covered only five attributes, and that a feedback-free classifier nearly matched the posterior on L. Before generating new data we committed an analysis plan (Nosek et al., 2018): labels, detectors (V6 primary), every comparison, the statistics, and the wording each outcome licenses. Seven amendments followed: three before any confirmatory data was seen (the supervised feedback increment, label-budget curves, the analyses of Section 7.3, H5's wording); the second domain, before any of its data existed; a replacement simulated patient, after the first failed its gate; V7 with H7 and H8, after the registered results and the intake-LR+F scores for four agents were known and before V7's first confirmatory run, when no detector had been run on the Llama-70B or travel corpora; and V7 as the question picker (H9), after the registered clarification test had failed and before V7's first run as picker. The confirmatory corpora use a fresh seed, show the agent the raw reaction, draw three resamples on every turn, and cover five agents (Llama-3.3-70B, Llama-3.1-8B, Qwen2.5-72B, GPT-4o-mini, Mistral-Small-3.1-24B), 600 sessions each (the Llama-70B run stopped at 375 sessions when API credit ran out and was

resumed with the same seed; no session was regenerated or dropped).

Statistics. Splits are 70/30 by session for three split seeds; detectors are fit on the training split and score the test split once; AUROC is averaged over seeds. Differences use a paired bootstrap (2,000 resamples) clustered by session, since test sets of different seeds overlap. Per-agent differences are computed on the test rows pooled over the three seeds, so they can differ in the third decimal from differences of the tables' seed-mean AUROCs. Pooled estimates average per-agent differences, with sessions resampled within agent. A difference is called significant when its 95% interval excludes zero; we do not correct for multiplicity, and intervals are conditional on the fitted detectors. Because the agents share sessions, we also resample sessions jointly across agents and report Holm-adjusted p-values (Table 4). The primary hypotheses are in Table 4, the per-agent comparisons in Appendix A, and every other registered comparison in the repository's analysis output.

Oracle-information analyses. Because the simulator's rules and hidden-field prior are known, we can compute three exact quantities. The *population risk* $\bar{\eta} = \sum_h P(h)\, L_U(C,h)$ is the probability that the session's commitments would be wrong for a random user with the same intake. $\eta_X = P(L_U = 1 \mid X, C)$ adds the reactions the agent was shown (all but the last); by Theorem 1 it bounds every detector computed from the agent's context and its resamples. η_F adds the last reaction and bounds every detector of commitments and reactions. The agent's policy does not involve the hidden facts, so its factors cancel and no agent model is needed. These are not deployable detectors; they locate the information. We also give V6 and ctx-LR(+F) the same oracle knowledge of which commitments are user-dependent in each session (V6-session, LR-session). (The registered plan calls $\bar{\eta}$, η_X and η_F eta0, etaX and etaF.)

7 Results

7.1 Why sampling detectors fail

Table 2 shows the mechanism. On commitments whose truth depends on undisclosed facts, resamples disagree about as often when the agent is right as when it is wrong: e_1 and e_0 differ by at most 0.07 on every agent, so by Corollary 1 the AUROC of disagreement is centred within half that of $\frac{1}{2}$. Llama-70B, Qwen-72B and GPT-4o-mini mostly repeat themselves; Llama-8B and Mistral-24B disagree with themselves often, but nearly as often when right as when wrong. One reaction from our simulated user gives 0.75, as designed; the comparison shows how little a user's reaction needs to carry to beat resampling, not how much a real user's does. At the session level (Table 3), SelfCheckGPT and semantic entropy score 0.551 [0.524, 0.578] and 0.495 [0.481, 0.508] pooled over agents, both intervals inside the registered band of $[0.40, 0.60]$ (H1): sampling detectors are near chance on user-specific hallucination. Resampling sessions jointly across agents, which share them, gives 0.551 [0.520, 0.585] and 0.495 [0.480, 0.509], both still inside the band. Per agent, SelfCheckGPT ranges over 0.492–0.610; the top value (Llama-8B) lies just outside the band. Self-verification does no better: Chain-of-Verification on Llama-70B scores 0.510. Sampling is not blind everywhere: on the composite label, where many errors are guesses about clinic rules that do vary across resamples, SelfCheckGPT reaches 0.645 [0.617, 0.672]. The failure is specific to facts the user never disclosed.

Table 2: Per-commitment check of Corollary 1 on commitments (any confidence) whose truth is user-dependent in that session. e_1, e_0: probability that some resample differs from the commitment, given that it is wrong / right. AUROC for predicting *that* commitment's error; the reaction column is on commitments that received one and equals $1 - \varepsilon = 0.75$ by the simulator's design.

Agent	wrong	all agree	e_1	e_0	Disagreement	One reaction
Llama-70B	52.4%	91.4%	0.087	0.085	0.501	0.748
Llama-8B	39.7%	49.5%	0.544	0.478	0.555	0.746
Qwen-72B	43.1%	83.1%	0.189	0.154	0.517	0.744
GPT-4o-mini	42.9%	85.6%	0.183	0.115	0.535	0.746
Mistral-24B	40.6%	69.2%	0.344	0.283	0.534	0.746

7.2 Detecting user-specific hallucination

Table 3 gives the confirmatory results and Table 4 the registered hypotheses, pooled over five agents.

Registered detectors. The user's implicit reactions add detectable signal beyond population base rates: +.070 [+.055, +.086]* to the posterior over its cell priors (H2; significant for 5 of 5 agents), and beyond the commitments themselves for the supervised detector (+.061 [+.041, +.080]*, H2b). Detectors on USHER's observables beat both sampling detectors (H4: +.112 [+.081, +.143]* and +.169 [+.139, +.198]*; H4b: +.162 [+.126, +.195]* and +.219 [+.190, +.243]*). Against the registered supervised detectors the first posterior, V6, did not win: the comparison with the commitment-only classifier is null (H3: +.011 [−.021, +.043]; registered wording: "context alone matches [USHER] on this label"), and ctx-LR+F is ahead with the full training split (−.050 [−.076, −.021]*). V3, with a symmetric channel, is indistinguishable from V6 (+.000 [−.004, +.003]).

The intake-aware posterior (Amendment 6). These results, and the oracle analysis of Section 7.3, located V6's weaknesses: its cells omit the intake, and its event is a proxy for the commitments the intake leaves to the user. V7 repairs both (Section 5) and reaches 0.825 [0.807, 0.843] AUROC pooled (Table 3). Conditioning on the intake improves the posterior (H7: +.162 [+.136, +.188]*),

Table 3: Session AUROC on the user-specific label L_U (mean of three splits; confirmatory corpora, 600 sessions per agent). †Added after the registered plan: ctx-LR+F plus the intake answers and commitment×intake cells; the comparator of H8 (Amendment 6). ‡Registered in Amendment 6 after the registered results for the other four agents were known; designed on development data only. CoVe was run for the 70B agent only. §GPT-4o-mini reading the whole transcript with its reactions (zero-shot; exploratory; on the GPT-4o-mini corpus it judges its own transcripts). The composite label L is in the repository.

	Detector	Llama-70B	Llama-8B	Qwen-72B	GPT-4o-mini	Mistral-24B
Baselines	SelfCheckGPT	.492	.610	.559	.576	.520
	Semantic entropy	.497	.533	.485	.486	.474
	CoVe	.510	–	–	–	–
	LLM judge§	.645	.611	.647	.612	.627
USHER: commitments	Mean confidence	.550	.512	.522	.465	.454
	ctx-LR	.671	.686	.715	.635	.556
	V6−prior	.634	.591	.593	.575	.577
USHER: commitments + reactions	Contradicted confident	.613	.614	.628	.576	.637
	Contradicted turns	.563	.536	.579	.595	.624
	ctx-LR+F	.745	.725	.748	.668	.682
	Intake-LR+F†	.743	.782	.749	.717	.690
	USHER posterior (V6)	.695	.672	.650	.636	.666
	USHER posterior, intake-aware (V7)‡	**.902**	**.889**	**.810**	**.836**	**.690**
	L_U positive rate	39.5%	29.7%	39.0%	35.3%	40.3%

and given the same information the posterior beats the supervised detector (H8: +.089 [+.067, +.114]*; significant for 4 of 5 agents; "same information" means the same raw inputs, while V7 also encodes the schema's structure). V7 − ctx-LR+F is +.112 [+.090, +.139]* (H8′); registered without claims, V7 − SelfCheckGPT and V7 − semantic entropy are +.274 [+.244, +.306]* and +.330 [+.308, +.354]*, and on L V7 reaches 0.924 [0.910, 0.935] with V7 − intake-LR+F +.027 [+.011, +.042]*. On Llama-70B, on which no detector had been run at registration, H7 is +.207* and H8 +.160*. The gain is smallest on Mistral-24B, where neither is significant (H7 +.027, H8 +.007).

Stronger supervised detectors on the same raw inputs (exploratory) still trail V7: with C chosen by cross-validation by +.070 [+.049, +.094]*, gradient-boosted trees by +.058 [+.039, +.082]*. A logistic regression handed V7's own structure (*structured LR+F*: cells conditioned on the intake answers each attribute depends on, schema-level attributes only, C by cross-validation) matches it (+.012 [−.005, +.033]): what wins is the structure, which commitments the intake leaves to the user, more than the estimator. The user's reactions carry part of it: scored without them, V7 loses +.083 [+.070, +.099]* (significant for 5 of 5 agents).

A zero-shot LLM judge that reads the whole transcript, reactions included (GPT-4o-mini; exploratory), reaches 0.628 [0.601, 0.656]: above SelfCheckGPT (+.077 [+.044, +.110]*) but far below V7 (+.197 [+.169, +.224]*, significant for 5 of 5 agents). Reading the reactions is not enough without the population structure.

Label budgets. Labels for user-specific hallucination require knowing the user's private facts, so small budgets are the realistic case (Figure 1; descriptive). V7's mean leads intake-LR+F at every budget (0.639 against 0.604 with 25 labels, 0.772 against 0.717 with 200). The classifier handed V7's structure is at least as good (0.648 and 0.776): with the structure given, the posterior's advantage is per-commitment risk, not session accuracy.

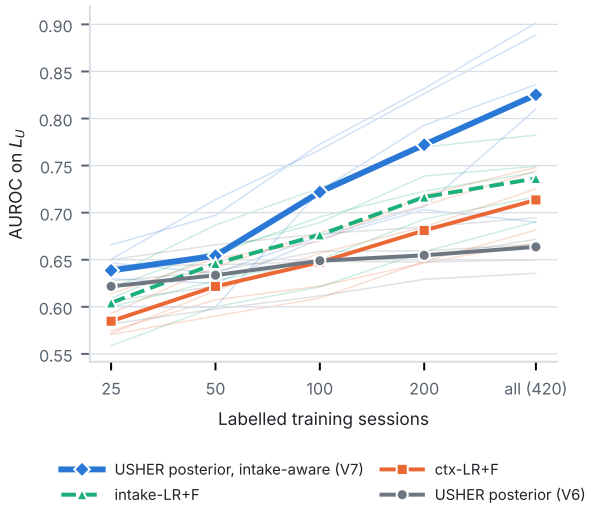


Figure 1: Test AUROC on L_U against the number of labelled training sessions (mean over agents; thin lines: individual agents; identical training subsets for every detector).

Table 4: Registered hypotheses on L_U, pooled over five agents (stratified session bootstrap; *: 95% CI excludes 0). Holm: Holm-adjusted p over the difference tests, with sessions resampled jointly across agents, which share them. H7, H8 and H8′ are Amendment 6.

	Comparison	Estimate [95% CI]	Holm
H1	AUROC(SelfCheckGPT)	0.551 [0.524, 0.578]	–
H1	AUROC(semantic entropy)	0.495 [0.481, 0.508]	–
H2	V6 − V6−prior	+.070 [+.055, +.086]*	<.001
H2b	ctx-LR+F − ctx-LR	+.061 [+.041, +.080]*	<.001
H3	V6 − ctx-LR	+.011 [−.021, +.043]	.555
H4	V6 − SelfCheckGPT	+.112 [+.081, +.143]*	<.001
H4	V6 − semantic entropy	+.169 [+.139, +.198]*	<.001
H4b	ctx-LR+F − SelfCheckGPT	+.162 [+.126, +.195]*	<.001
H4b	ctx-LR+F − semantic entropy	+.219 [+.190, +.243]*	<.001
–	V6 − ctx-LR+F	−.050 [−.076, −.021]*	.012
H5	$\eta_F - \bar{\eta}$ (oracle)	+.083 [+.068, +.100]*	<.001
H6	V6-session − LR-session+F (oracle)	+.019 [+.006, +.037]*	.136
H7	V7 − V6	+.162 [+.136, +.188]*	<.001
H8	V7 − intake-LR+F	+.089 [+.067, +.114]*	<.001
H8′	V7 − ctx-LR+F	+.112 [+.090, +.139]*	<.001

Table 5: Oracle-information analysis on L_U (AUROC). $\bar{\eta}$: population risk of the commitments; η_X: Bayes given the conversation the agent saw (Theorem 1's bound for context-and-sample detectors); η_F: Bayes given all reactions. V6-session and LR-session+F are told which commitments are user-dependent in each session. Pooled differences are in Table 4.

Agent	$\bar{\eta}$	η_X	η_F	V6-sess.	LR-sess.+F
Llama-70B	.876	.942	.944	.913	.885
Llama-8B	.899	.952	.953	.904	.908
Qwen-72B	.854	.930	.932	.879	.851
GPT-4o-mini	.803	.907	.909	.837	.825
Mistral-24B	.781	.890	.891	.847	.814

7.3 Where the information is

Table 5 locates the signal. The population risk $\bar{\eta}$ alone is far above every sampling detector: much of L_U is visible in *which* values an agent commits to, given the intake. The user's reactions add $\eta_F - \bar{\eta}$ beyond it (H5). Almost all of that information reaches the agent's transcript ($\eta_F - \eta_X$ is +.001 [−.001, +.004]): reactions to earlier commitments are displayed later in the session, so a detector reading the transcript could use them. Per-turn resampling cannot: agreement between samples ignores the reactions, and each resample is drawn before its own commitment's reaction. Given the same information (which commitments concern undisclosed facts), the USHER posterior beats the supervised detector (H6), by a small margin (+.019 [+.006, +.037]*) that does not survive resampling sessions jointly across agents (Holm $p = .136$), so we do not rely on it. Without that knowledge, V6 targets a schema-level set of attributes that is only a proxy for the label's support, which costs it 0.181–0.233 AUROC against V6-session, an order of magnitude more than the H6 margin: knowing *which* commitments concern undisclosed facts matters far more than how the evidence is combined. A deployable detector has to learn this from labels. A generic classifier mostly does not (intake-LR+F reaches 0.690–0.782, against 0.781–0.899 for the population risk computed with the rules). V7, which learns it as the posterior's label event, closes 76% of V6's gap to V6-session (mean 0.876) without oracle knowledge.

Table 6: Second domain (travel booking), AUROC on L_U, 600 sessions per agent. No detector had been run on these corpora when V7 was registered.

Agent	SelfCheck	ctx-LR+F	intake-LR+F	V6	V7	$\bar{\eta}$	η_F
GPT-4o-mini	.535	.658	.684	.682	**.756**	.786	.862
Llama-8B	.628	.825	.877	.796	**.904**	.944	.963

7.4 A second domain

In travel booking (Table 6), V7, run once on each corpus, reaches 0.830 [0.795, 0.855] pooled over the two agents. Within this domain, as registered, conditioning on the intake improves the posterior (H7: +.091 [+.054, +.117]*), and given the same information the posterior beats the supervised detector (H8: +.050 [+.010, +.084]*; H8′: +.089 [+.046, +.126]*), so H7 and H8 hold in each domain separately. Per agent, H8 is significant for GPT-4o-mini (+.067 [+.008, +.128]*) but not for Llama-8B (+.028 [−.014, +.073]). V7 beats SelfCheckGPT by +.249 [+.197, +.297]*. Sampling is less blind for Llama-8B: its resamples disagree with 0.512 of its wrong user-dependent commitments and 0.230 of its right ones, so by Corollary 1 disagreement should score near $\frac{1}{2} + \frac{1}{2}(e_1 - e_0)$; it scores 0.651, and SelfCheckGPT 0.628, above H1's band. Pooled over the two agents SelfCheckGPT scores 0.582 [0.533, 0.626], whose interval extends above the band, so in this domain we do not call it near chance (semantic entropy: 0.507 [0.499, 0.520]). The other registered tests replicate within the domain (H2 +.079 [+.049, +.118]*, H2b +.045 [+.010, +.083]*, H4 and H4b +.157 to +.234, all significant, H5 +.048 [+.025, +.073]*), and H3 and H6 are null, as in healthcare. For this agent the intake fixes more of the risk: the population risk alone reaches 0.944, and the reactions add +.021 [+.002, +.041]* to it.

7.5 Natural-language reactions

The first simulated patient failed its gate. Among commitments it replied to, it contradicted a fraction 0.83 of the wrong ones and 0.79 of the correct ones (lift 1.06; 100 sessions), against 0.75 and 0.25 for the noisy bit; the informed variant, told the clinic's decision, did little better (0.97 and 0.80, lift 1.21), and on the pilot its reactions added nothing detectable to V6 (V6 − V6−prior +.017 [−.048, +.084]). Reading the replies shows why: the patient voiced doubt about facts it had not been given, and the labeller counted doubt as contradiction. As registered, neither was run at scale. A second version, registered before its data existed (Amendment 5), uses GPT-4o-mini for both roles, lets the patient object only to commitments that conflict with its

listed facts, and labels a reply as a contradiction only if it states a conflict. The second version passed: on the pilot it contradicted 0.15 of the wrong commitments it replied to and 0.06 of the correct ones (lift 2.69), so all 600 sessions were run. Its reactions are much weaker than the noisy bit's: it is silent after 0.63 of wrong commitments and accepts 0.22 of them. In the registered analysis (without claims) its reactions add +.051 [+.023, +.076]* to the posterior over its priors, but ctx-LR, which ignores them, scores 0.729 against 0.644 for V6, and adding them to it changes nothing detectable (+.005 [−.034, +.043]). The frozen V7 (exploratory here) scores 0.719, below intake-LR+F (−.060 [−.115, −.003]*): this patient's reactions vary by attribute, which V7's shared channel does not model. With realistic reactions, most of the detectable risk is in the commitments themselves.

7.6 Choosing what to ask

The posterior's per-commitment risk q_t is what chooses the question, and no session-level classifier provides it. On confident commitments whose truth is user-dependent in that session (registered in Amendment 6, without a claim), V7's q_t identifies the wrong ones with AUROC 0.797 [0.768, 0.825] (per agent 0.766–0.834) and calibration error 0.081, against 0.752 [0.715, 0.787] for the commitment's own reaction, 0.541 [0.521, 0.560] for resample disagreement and 0.521 [0.492, 0.550] for stated confidence (per split seed, session-clustered intervals). Over all confident commitments on the schema-level attributes, a set a deployed system can know (exploratory), it reaches 0.892 [0.879, 0.906], against 0.739 [0.713, 0.764], 0.568 [0.557, 0.579] and 0.629 [0.610, 0.648]. The measured loop (70B agent) uses the first 120 test sessions of split seed 42 instead of the registered 180, a deviation forced by the API budget. The registered test failed. On the primary outcome, wrong confident user-dependent commitments per session (0.642 without questions), one question chosen by V6's risk did no better than one chosen at random (−.008 [−.133, +.117]), so the registered claim that choosing by posterior risk beats choosing at random is not supported, and it was not detectably better than rerunning without a question (−.125 [−.242, +.000]). The registered picker ranked commitments by V6 fit to the composite label, which counts rule errors. Asking about every user-dependent attribute (4.29 questions per session) leaves far fewer such errors than two posterior-chosen questions (+.433 [+.325, +.550]*). On the composite label the two are not distinguishable (−.017 [−.050, +.017]), but there the simulated patient answers questions about rule-determined attributes truthfully, which a real patient could not. With V7, fit to L_U, as the picker (Amendment 7; same sessions, patient and noise), the test succeeds: choosing the question by V7's per-commitment risk beats choosing at random. One V7-chosen question leaves 0.367 wrong confident user-dependent commitments per session, against 0.583 for a random one (H9: −.217 [−.325, −.108]*); it also beats the registered V6 picker (−.208 [−.308, −.108]*) and rerunning without a question (−.333 [−.458, −.208]*). Two V7-chosen questions leave 0.333; asking about every user-dependent attribute, with 4.29 questions, leaves 0.108.

7.7 Robustness to the user's behaviour

Our users react with a designed noise level. To see how much that matters, we regenerated the reactions seen by the detectors on the confirmatory trajectories, with the agents' trajectories fixed (Theorem 2's setting; exploratory; mean over agents of 5 draws). As the flip rate rises from 0.05 to 0.45, V7's mean AUROC on L_U falls from 0.961 to 0.669, faster than the Bayes predictor given the intake, the commitments and the same regenerated reactions (0.989 to 0.847), against 0.883 to 0.685 for intake-LR+F. With reactions to half or a quarter of the commitments, V7 scores 0.766 and 0.693 (intake-LR+F: 0.704 and 0.688). Self-CheckGPT, which reads no reactions, stays at 0.551. This measures the detectors' robustness only: the agents conditioned on the recorded reactions.

8 Limitations

Simulation. The agents are real but the user is simulated. In the main corpora it knows whether each commitment is correct with probability 0.75; real users know their facts, not the clinic's rules, which the LLM patient tests; with its weaker reactions V7 loses its advantage (Section 7.5). The oracle analyses rely on the simulator's known rules and prior, which a deployment does not have. *Two synthetic domains.* Both private-attribute domains are simulators we designed, with deterministic rules and independent hidden facts; real user populations have correlated, partially self-reported facts. *Post-hoc method.* V7 was designed after the registered results were known, tuned on development data only, and registered before its first confirmatory run; Llama-70B and the travel domain are the corpora on which no detector had been run when it was registered. *Labels.* Labels are programmatic and noise-free by construction; there is no human evaluation. *Supervision.* The posterior and the supervised baselines are fit on labelled training sessions; the posterior can run without labels when the channel is known, which we only test in development. *Untested detectors.* Our LLM judge is zero-shot; a judge trained on labelled transcripts also reads the reactions and is not bounded by Theorem 1. Sampling detectors used three resamples (ten were registered as optional and not run), and with exact matching semantic entropy takes few values. We did not run hidden-state probes; Theorem 1 predicts they share the within-context limit, but they could exploit population risk as ctx-LR does. *Theory.* Proposition 1 is exact only under (A1)–(A2), and our detectors are plug-in estimates; Theorem 2 holds only with the agent's trajectories fixed.

9 Conclusion

When an agent invents a fact about its user, its own samples cannot tell which of the users sharing its context it is wrong about. Some of the risk is visible in what the agent commits to; the rest is visible only through the user's reactions. Making the agent state its assumptions as probabilistic commitments and recording the user's reactions makes both available, and a Bayes analysis in a simulator with known truth shows how much each is worth. An intake-aware posterior turns them into session and per-commitment risk; the structure it encodes, which commitments the intake leaves to the user, is what separates the detectors we tested. Confirming the commitments its risk ranks highest removes more user-specific errors in a closed loop than random questions.

Reproducibility. Code, all trajectories, the pre-registered plan with its seven amendments and a log of their commit times, every analysis script, and a ledger of all runs including discarded variants are at [PENDING: anonymous repository link].

Use of AI tools. AI coding and writing assistants were used to implement experiments and analysis code and to draft and edit text. All reported numbers are produced by the released code; the authors take full responsibility for the content.

References

Abbasi Yadkori, Y., Kuzborskij, I., György, A., and Szepesvári, C. (2024). To believe or not to believe your LLM: Iterative prompting for estimating epistemic uncertainty. In *NeurIPS*, volume 37.

Angluin, D. and Laird, P. (1988). Learning from noisy examples. *Machine Learning*, 2.

Azaria, A. and Mitchell, T. (2023). The internal state of an LLM knows when it's lying. In *Findings of EMNLP*.

Blackwell, D. (1953). Equivalent comparisons of experiments. *Ann. Math. Statist.*, 24.

Bohus, D. and Rudnicky, A. I. (2005). Constructing accurate beliefs in spoken dialog systems. In *IEEE ASRU*.

Chen, C., Liu, K., Chen, Z., Gu, Y., Wu, Y., Tao, M., Fu, Z., and Ye, J. (2024). INSIDE: LLMs' internal states retain the power of hallucination detection. In *ICLR*.

Clark, H. H. and Brennan, S. E. (1991). Grounding in communication. In *Perspectives on Socially Shared Cognition*. APA.

Corbett, A. T. and Anderson, J. R. (1994). Knowledge tracing: Modeling the acquisition of procedural knowledge. *User Model. User-Adapt. Interact.*, 4.

Dawid, A. P. and Skene, A. M. (1979). Maximum likelihood estimation of observer error-rates using the EM algorithm. *J. R. Stat. Soc. C*, 28.

Deng, M., Li, Z., Li, X., Zhu, T., Zhao, Y., Guo, Z., and Wang, W. (2026). Uncertainty-aware clarification in LLM agents with information gain. In *ICML*.

Dhuliawala, S., Komeili, M., Xu, J., Raileanu, R., Li, X., Celikyilmaz, A., and Weston, J. (2024). Chain-of-verification reduces hallucination in large language models. In *Findings of ACL*.

Dziri, N., Milton, S., Yu, M., Zaiane, O., and Reddy, S. (2022). On the origin of hallucinations in conversational models: Is it the datasets or the models? In *NAACL*.

Farquhar, S., Kossen, J., Kuhn, L., and Gal, Y. (2024). Detecting hallucinations in large language models using semantic entropy. *Nature*, 630.

Hancock, B., Bordes, A., Mazaré, P.-E., and Weston, J. (2019). Learning from dialogue after deployment: Feed yourself, chatbot! In *ACL*.

Hoeffding, W. (1963). Probability inequalities for sums of bounded random variables. *J. Amer. Statist. Assoc.*, 58.

Hu, Z., Liu, C., Feng, X., Zhao, Y., Ng, S.-K., Luu, A. T., He, J., Koh, P. W., and Hooi, B. (2024). Uncertainty of thoughts: Uncertainty-aware planning enhances information seeking in large language models. In *NeurIPS*, volume 37.

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y. J., Madotto, A., and Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Comput. Surv.*, 55.

Joachims, T., Granka, L., Pan, B., Hembrooke, H., and Gay, G. (2005). Accurately interpreting clickthrough data as implicit feedback. In *SIGIR*.

Kadavath, S. et al. (2022). Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*.

Kalai, A. T. and Vempala, S. S. (2024). Calibrated language models must hallucinate. In *STOC*.

Kipf, T. N. and Welling, M. (2017). Semi-supervised classification with graph convolutional networks. In *ICLR*.

Kirmayr, J., Stappen, L., and André, E. (2026). CAR-bench: Evaluating the consistency and limit-awareness of LLM agents under real-world uncertainty. *arXiv preprint arXiv:2601.22027*.

Kossen, J., Han, J., Razzak, M., Schut, L., Malik, S., and Gal, Y. (2024). Semantic entropy probes: Robust and cheap hallucination detection in LLMs. *arXiv preprint arXiv:2406.15927*.

Kuhn, L., Gal, Y., and Farquhar, S. (2023). Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *ICLR*.

Li, S. S., Balachandran, V., Feng, S., Ilgen, J. S., Pierson, E., Koh, P. W., and Tsvetkov, Y. (2024). MediQ: Question-asking LLMs and a benchmark for reliable interactive clinical reasoning. In *NeurIPS*, volume 37.

Liu, X., Yang, X., Li, Z., Li, P., and He, R. (2026). AgentHallu: Benchmarking automated hallucination attribution of LLM-based agents. *arXiv preprint arXiv:2601.06818*.

Lovász, L. (2012). *Large Networks and Graph Limits*. AMS.

Manakul, P., Liusie, A., and Gales, M. J. F. (2023). SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models. In *EMNLP*.

Mündler, N., He, J., Jenko, S., and Vechev, M. (2024). Self-contradictory hallucinations of large language models: Evaluation, detection and mitigation. In *ICLR*.

Natarajan, N., Dhillon, I. S., Ravikumar, P., and Tewari, A. (2013). Learning with noisy labels. In *NeurIPS*.

Neyman, J. and Pearson, E. S. (1933). On the problem of the most efficient tests of statistical hypotheses. *Phil. Trans. R. Soc. A*, 231.

Nosek, B. A., Ebersole, C. R., DeHaven, A. C., and Mellor, D. T. (2018). The preregistration revolution. *PNAS*, 115.

Rao, S. and Daumé III, H. (2018). Learning to ask good questions: Ranking clarification questions using neural expected value of perfect information. In *ACL*.

Ren, A. Z., Dixit, A., Bodrova, A., Singh, S., Tu, S., Brown, N., Xu, P., Takayama, L., Xia, F., Varley, J., Xu, Z., Sadigh, D., Zeng, A., and Majumdar, A. (2023). Robots that ask for help: Uncertainty alignment for large language model planners. In *CoRL*.

Schatzmann, J., Weilhammer, K., Stuttle, M., and Young, S. (2006). A survey of statistical user simulation techniques for reinforcement-learning of dialogue management strategies. *Knowl. Eng. Rev.*, 21.

Scott, C., Blanchard, G., and Handy, G. (2013). Classification with asymmetric label noise: Consistency and maximal denoising. In *COLT*.

Sharma, M., Tong, M., Korbak, T., et al. (2024). Towards understanding sycophancy in language models. In *ICLR*.

Singh, M. P. (1999). An ontology for commitments in multiagent systems: Toward a unification of normative concepts. *Artif. Intell. Law*, 7.

Suri, M., Mathur, P., Lipka, N., Dernoncourt, F., Rossi, R. A., and Manocha, D. (2025). Structured uncertainty guided clarification for LLM agents. *arXiv preprint arXiv:2511.08798*.

Tian, K., Mitchell, E., Zhou, A., Sharma, A., Rafailov, R., Yao, H., Finn, C., and Manning, C. D. (2023). Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *EMNLP*.

Ulmer, D., Gubri, M., Lee, H., Yun, S., and Oh, S. J. (2024). Calibrating large language models using their generations only. In *ACL*.

Wei, J., Yang, C., Song, X., Lu, Y., Hu, N., Huang, J., Tran, D., Peng, D., Liu, R., Huang, D., Du, C., and Le, Q. V. (2024). Long-form factuality in large language models. In *NeurIPS*.

Williams, J. D. and Young, S. (2007). Partially observable Markov decision processes for spoken dialog systems. *Comput. Speech Lang.*, 21.

Xiong, M., Hu, Z., Lu, X., Li, Y., Fu, J., He, J., and Hooi, B. (2024). Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. In *ICLR*.

Yao, S., Shinn, N., Razavi, P., and Narasimhan, K. (2024). τ-bench: A benchmark for tool-agent-user interaction in real-world domains. *arXiv preprint arXiv:2406.12045*.

Zhang, M., Press, O., Merrill, W., Liu, A., and Smith, N. A. (2024). How language model hallucinations can snowball. In *ICML*.

Zhang, M. J. Q. and Choi, E. (2025). Clarify when necessary: Resolving ambiguity through interaction with LMs. In *Findings of NAACL*.

Zhang, Q., Durfee, E., Singh, S., Chen, A., and Witwicki, S. (2016). Commitment semantics for sequential decision making under reward uncertainty. In *IJCAI*.

Zhang, W., Sun, Y., Huang, P., Pu, J., Lin, H., and Song, D. (2025). MIRAGE-Bench: LLM agent is hallucinating and where to find them. *arXiv preprint arXiv:2507.21017*.

A Per-agent registered comparisons

Table 7 gives every registered difference per agent on L_U, and pooled on L (point estimates; *: the paired session-clustered 95% CI excludes 0; intervals in the repository). H2: V6 − V6−prior; H2b: ctx-LR+F − ctx-LR; H3: V6 − ctx-LR; H4: V6 − SelfCheckGPT (SC) and semantic entropy (SE); H7: V7 − V6; H8: V7 − intake-LR+F.

Table 7: Registered differences per agent (AUROC).

Agent	H2	H2b	H3	H4 (SC)	H4 (SE)	V6−ctx-LR+F	H7	H8
User-specific label L_U (primary)								
Llama-70B	+.060*	+.070*	+.026	+.201*	+.197*	−.043	+.207*	+.160*
Llama-8B	+.079*	+.040*	−.016	+.060	+.139*	−.056*	+.217*	+.108*
Qwen-72B	+.057*	+.033	−.070*	+.086*	+.159*	−.103*	+.163*	+.060*
GPT-4o-mini	+.060*	+.034	+.005	+.058*	+.151*	−.029	+.198*	+.120*
Mistral-24B	+.090*	+.126*	+.115*	+.150*	+.197*	−.011	+.027	+.007
Pooled	+.070*	+.061*	+.011	+.112*	+.169*	−.050*	+.162*	+.089*
Composite label L (secondary; per agent in the repository)								
Pooled	+.073*	+.028*	+.041*	+.209*	+.306*	+.013	+.070*	+.027*